

Expand on claims paper

 2310.17813v2.pdf
PDF

Hello, I would like you to expand on the arguments given for the claims in a paper I am reading (A Spectral Condition ...). When doing so, could you please make sure your explanation is:

- A. Precise
- B. Easy to follow: proofs should be in English prose as much as possible (while still being precise)
- C. Self-Contained: Please try to only reference facts a CS182 student would know. If you need to include new facts/arguments, please state them in full before using them
- D. Well structured

Also, at the end, please try to generate some questions to check conceptual understanding

Expansion and clean proofs of the two *alignment* claims (plain English, self-contained)

Thanks — I read the excerpt you uploaded from *A Spectral Condition for Feature Learning* (the PDF you gave). 

Below I rewrite and expand the arguments for the two claims the authors state (their Claim 1 and Claim 2), giving precise statements, the elementary facts used, short proofs written mainly in English, and the way those claims lead to the bounds in Equations (8)–(10). I keep the math at the level of a typical CS182 course (linear algebra + basic probability / concentration). At the end I give conceptual check questions.

Roadmap / notation (so this is self-contained)

- Layer index: I write a single hidden layer as $W \in \mathbb{R}^{m \times n}$ where n is the input dimension to the layer (fan-in) and m is the output dimension (fan-out). The input feature to the layer is a vector $h_{\text{in}} \in \mathbb{R}^n$ and the post-multiplication feature is $h_{\text{out}} = Wh_{\text{in}} \in \mathbb{R}^m$.
- Vector / operator norms: $\|v\|_2$ is the Euclidean norm of a vector. The spectral/operator norm of a matrix A is $\|A\|_2 = \sup_{\|x\|_2=1} \|Ax\|_2$. For clarity I will write $\|A\|_{\text{op}}$ when I mean the spectral norm.
 ← technically true but could have also said $\max_i (\|Ax_i\|_2)$, where $x_1, \dots, x_n$ is the standard $\mathbb{R}^n$ basis
- “ $\asymp$ ” or the paper’s “!” notation: the authors use notation like $\|Wh\| \approx \|W\|_{\text{op}} \|h\|$. I will quantify this: sometimes we will get an exact equality (rank-one updates), sometimes a high-probability approximate equality with explicit multiplicative constants.
 ← Not sure where this “op” came from I think it refers to “operator” even though W is rectangular...
- Elementary fact (used repeatedly): for an outer product ab^T (with $a \in \mathbb{R}^m$, $b \in \mathbb{R}^n$) the operator norm equals $\|ab^T\|_{\text{op}} = \|a\|_2 \|b\|_2$. The top singular vector is in the directions of a and b . (Proof: singular values of ab^T are $\|a\| \|b\|$ and zeros; this is standard linear algebra.)
- Probability facts I will use (stated before use):

1. **Gaussian-vector scaling:** If $w \in \mathbb{R}^m$ has i.i.d. entries $\mathcal{N}(0, \sigma^2)$ and $u \in \mathbb{R}^n$ is fixed, then Wu has i.i.d. Gaussian entries with variance $\sigma^2 \|u\|_2^2$. Hence $\mathbb{E} \|Wu\|_2^2 = m\sigma^2 \|u\|_2^2$, and by concentration of the χ^2 distribution (or a simple law of large numbers argument) $\|Wu\|_2 \approx \sigma\sqrt{m} \|u\|_2$ with high probability for large m . ← this is easy to prove, it would have been better to show this

2. **Spectral norm of a Gaussian matrix (Rudelson-Vershynin / Marchenko-Pastur intuition):** If $W \in \mathbb{R}^{m \times n}$ has i.i.d. $\mathcal{N}(0, \sigma^2)$ entries then with high probability

$$\|W\|_{\text{op}} \approx \sigma(\sqrt{m} + \sqrt{n}),$$

← this is also easy to prove, it would have been better to show this

and for many regimes $\|W\|_{\text{op}}$ concentrates around that value. This is a standard nonasymptotic result in random matrix theory (see e.g. Vershynin notes). I will use this as a black-box estimate; the important qualitative implication is that $\|W\|_{\text{op}}$ is on the order of $\sigma\sqrt{m}$ when $m \geq n$ (more precisely it is at most a constant factor larger when m and n are comparable).

Claim 1 (alignment of *initial* random weight matrices) — precise restatement

← the proof here now makes sense since it is really just rearranging what it's already told us. I feel like it doesn't help that much with intuition though because the meat of the proof is hidden behind "facts" that were just cited

Claim 1 (precise). Let $W \in \mathbb{R}^{m \times n}$ be an initialization for a layer with fan-out m and fan-in n . Suppose either

1. W is *semi-orthogonal* with $m \geq n$ in the sense $W^\top W = I_n$ (columns orthonormal), or
2. W has i.i.d. entries $\mathcal{N}(0, \sigma^2)$ and m is not smaller than n (so $m \geq n$).

Then for any fixed input vector $h_{\text{in}} \in \mathbb{R}^n$ we have, with high probability in case (2),

$$\|Wh_{\text{in}}\|_2 \text{ is on the same scale as } \|W\|_{\text{op}} \|h_{\text{in}}\|_2,$$

more concretely:

- In case (1) exact equality holds: $\|Wh_{\text{in}}\|_2 = \|W\|_{\text{op}} \|h_{\text{in}}\|_2$ because $\|W\|_{\text{op}} = 1$ and $\|Wh\| = \|h\|$.
- In case (2) with Gaussian entries,

$$\|Wh_{\text{in}}\|_2 \approx \sigma\sqrt{m} \|h_{\text{in}}\|_2, \quad \|W\|_{\text{op}} \approx \sigma(\sqrt{m} + \sqrt{n}),$$

so whenever m is at least as large as n these two quantities agree up to a multiplicative constant bounded away from 0; in particular

$$\frac{\|Wh_{\text{in}}\|_2}{\|W\|_{\text{op}} \|h_{\text{in}}\|_2} \gtrsim \frac{\sqrt{m}}{\sqrt{m} + \sqrt{n}} \geq \frac{1}{2},$$

so $\|Wh_{\text{in}}\|_2$ and $\|W\|_{\text{op}} \|h_{\text{in}}\|_2$ are the same order when $m \geq n$.

Proof / explanation (in words) ← this is a more expanded version of the proof above

- **Semi-orthogonal case (exact):** If $W^\top W = I_n$ then all singular values of W equal 1. Thus $\|W\|_{\text{op}} = 1$. Also for any h , $\|Wh\|_2^2 = h^\top W^\top W h = h^\top h = \|h\|_2^2$. Hence $\|Wh\|_2 = \|h\|_2 = \|W\|_{\text{op}} \|h\|_2$. This is the easiest case and gives exact alignment.
- **Gaussian case (approximate):** Fix $h \neq 0$. Consider the random vector Wh . Each component $(Wh)_i$ is a linear combination of the n i.i.d. Gaussians in the i -th row of W . Because rows are independent and mean zero, $(Wh)_i$ is Gaussian with mean 0 and variance $\sigma^2 \|h\|_2^2$. Thus

$$\|Wh\|_2^2 = \sum_{i=1}^m (Wh)_i^2 \quad \text{is } \sigma^2 \|h\|_2^2 \cdot \chi_m^2.$$

The χ_m^2 random variable concentrates tightly around m (by the law of large numbers / standard concentration), so with high probability $\|Wh\|_2 \approx \sigma \sqrt{m} \|h\|_2$.

On the other hand, standard random matrix results give $\|W\|_{\text{op}} \approx \sigma(\sqrt{m} + \sqrt{n})$. When $m \geq n$ the term $\sqrt{m}$ dominates or is comparable to $\sqrt{n}$, so the ratio

$$\frac{\|Wh\|_2}{\|W\|_{\text{op}} \|h\|_2} \approx \frac{\sigma \sqrt{m}}{\sigma(\sqrt{m} + \sqrt{n})} = \frac{\sqrt{m}}{\sqrt{m} + \sqrt{n}}$$

Here \approx means as you consider everything in the limit

is bounded below by $1/2$. Thus $\|Wh\|_2$ is of the same order as $\|W\|_{\text{op}} \|h\|_2$. Intuitively: the spectral norm measures the *largest* singular value (how much one particular direction can be amplified). For a randomly oriented input vector, only the top singular direction could produce much smaller amplification — but for wide matrices (large m relative to n) the top singular value is not vastly bigger than the typical singular values, and a random h will typically pick up the bulk of the energy, so $\|Wh\|$ scales like $\|W\|_{\text{op}} \|h\|$ up to constants.

Takeaway. Semi-orthogonal initialization gives exact equality. Gaussian initialization gives the same scaling with high probability provided the layer is not narrow (fan-out $\geq$ fan-in).

Claim 2 (alignment of gradient/updates) — precise restatement and proof

Same commentary I said earlier applies here

Claim 2 (precise). Consider a gradient update at a single training step with batch size 1 at some layer with input h_{in} . Denote the weight update by ΔW . Under standard backpropagation, ΔW has the form (up to the learning rate scalar) of an outer product:

$$\Delta W = \eta a h_{\text{in}}^\top$$

for some vector a (this is the gradient flowing back multiplied by upstream factors), and scalar η is the learning rate. Then for that same input h_{in} we have the **exact equality**

$$\|\Delta W h_{\text{in}}\|_2 = \|\Delta W\|_{\text{op}} \|h_{\text{in}}\|_2.$$

Proof / explanation (in words)

- The gradient update ΔW for a single example is a rank-one matrix because the gradient w.r.t. a weight matrix in a fully connected layer is always an outer product of the backpropagated error

vector (call it a) and the input feature h_{in} . Concretely,

$$\Delta W = \eta a h_{\text{in}}^\top.$$

- By the elementary fact above, the operator norm of the rank-one matrix ΔW equals $\|\Delta W\|_{\text{op}} = \eta \|a\|_2 \|h_{\text{in}}\|_2$.
- Evaluate $\Delta W h_{\text{in}} = \eta a (h_{\text{in}}^\top h_{\text{in}}) = \eta \|h_{\text{in}}\|_2^2 a$. Thus

$$\|\Delta W h_{\text{in}}\|_2 = \eta \|h_{\text{in}}\|_2^2 \|a\|_2 = (\eta \|a\|_2 \|h_{\text{in}}\|_2) \|h_{\text{in}}\|_2 = \|\Delta W\|_{\text{op}} \|h_{\text{in}}\|_2.$$

- Everything above is exact — no randomness or approximation is needed. That explains why the authors state Claim 2 as an equality.

Takeaway. Rank-one structure of single-example gradient updates gives exact alignment of the update with the incoming feature — hence equality.

How the two claims give the scaling results (Equations (8)–(10) in the excerpt) I feel like this is unnecessary

The authors want to show that feature norms at initialization and feature-increment norms from a single SGD update scale in expected ways. I restate and justify their steps.

(A) Feature scale at initialization

A forward feature at layer ℓ is $h_\ell = W_\ell h_{\ell-1}$. Using Claim 1 (alignment of initialization) we have (up to constants in Gaussian init; exact in semi-orthogonal init)

$$\|h_\ell\|_2 = \|W_\ell h_{\ell-1}\|_2 \approx \|W_\ell\|_{\text{op}} \|h_{\ell-1}\|_2.$$

If the network is initialized so that $\|W_\ell\|_{\text{op}}$ has a prescribed scale (the authors call this Condition 1), then these norms propagate layer to layer multiplicatively. For a single layer the authors insert the Condition 1 that $\|W_\ell\|_{\text{op}}$ scales like $\sqrt{n_\ell}$ (or whatever normalization they use), giving the line labeled (8):

$$\|h_\ell(x)\|_2 \approx \|W_\ell\|_{\text{op}} \|h_{\ell-1}(x)\|_2 = \Theta(\sqrt{n_\ell}) \quad (\text{under the stated initialization condition}).$$

The important point is that Claim 1 ensures a *lower bound* (the vector is not projected entirely onto tiny singular directions) so the forward norm actually grows at the predicted spectral scale.

(B) Size of the single-step change in hidden feature Δh_ℓ

From first-order Taylor / the network update rule, the change in the hidden feature at layer ℓ after a small weight change ΔW_ℓ can be written (to first order) as

$$\Delta h_\ell = \Delta W_\ell h_{\ell-1} + W_\ell \Delta h_{\ell-1} + (\text{higher order term}) \Delta W_\ell \Delta h_{\ell-1}.$$

This is exactly the expansion the authors wrote as Equation (9).

- The first term $\Delta W_\ell h_{\ell-1}$ is the direct effect of the weight change on the current layer's output for the fixed input x . By Claim 2 we have the **exact** equality

$$\|\Delta W_\ell h_{\ell-1}\|_2 = \|\Delta W_\ell\|_{\text{op}} \|h_{\ell-1}\|_2.$$

So this term scales like $\|\Delta W_\ell\|_{\text{op}} \|h_{\ell-1}\|_2$.

- The other terms $W_\ell \Delta h_{\ell-1}$ and $\Delta W_\ell \Delta h_{\ell-1}$ are higher-order (the second is quadratic in small updates) or propagate changes from earlier layers. The authors assume these do not perfectly cancel the first term; under that generic (non-pathological) condition the magnitude of Δh_ℓ is comparable to the first term.

Putting these together and inserting their Condition 1 (which fixes the scale of $\|W_\ell\|_{\text{op}}$ and the scale of $\|\Delta W_\ell\|_{\text{op}}$ per update), they conclude (Equation (10)) that

$$\|\Delta h_\ell\|_2 = \Theta(\|\Delta W_\ell\|_{\text{op}} \|h_{\ell-1}\|_2) = \Theta(\sqrt{n_\ell}).$$

So both the forward features and their single-step changes scale like the spectral norm times the input magnitude; Claim 1 handles the forward pass, Claim 2 handles the single-step update.

A few remarks about the assumptions and constants (practical interpretation)

1. **Role of width (fan-out) m vs. fan-in n .** The Gaussian argument relies on m not being much smaller than n . If the layer is *narrow* (small m), then $\|W\|_{\text{op}}$ may be dominated by $\sqrt{n}$ and $\|Wh\|$ might be substantially smaller than $\|W\|_{\text{op}} \|h\|$ for some h . Semi-orthogonal inits are a clean fix because they give exact equality independent of dimensions.
2. **Concentration caveat.** All Gaussian statements are “with high probability” and rely on concentration inequalities. For practical finite widths the constants matter, but the asymptotic message is robust: wide layers behave predictably.
3. **Rank-one update exactness.** Because single-example SGD produces rank-one updates, Claim 2 is exact. For mini-batch updates the update is a sum of outer products (one per example in the batch). If the batch has size B then ΔW is at most rank B , and the equality in Claim 2 no longer strictly holds, but the same intuition remains: ΔWh will be in the span of the backpropagated vectors and again typically scales with $\|\Delta W\|_{\text{op}} \|h\|$ unless pathological cancellation occurs.
4. **“Non-perfect cancellation” assumption.** When the authors say “so long as the first term does not perfectly cancel with the latter two,” they mean: unless the backprop signals and existing weights line up in an adversarial way that exactly cancels magnitudes, the first term gives the dominant contribution. In practice this is a mild (and standard) genericity assumption.

Short summary (one paragraph)

- If a layer is initialized semi-orthogonally then the layer *preserves* vector norms exactly and hence $\|Wh\| = \|W\|_{\text{op}} \|h\|$. If it's Gaussian with reasonable width, then with high probability $\|Wh\|$ concentrates around $\sigma\sqrt{m}\|h\|$ while $\|W\|_{\text{op}}$ concentrates around $\sigma(\sqrt{m} + \sqrt{n})$: these two values are the same order when $m \geq n$. Thus random initial matrices typically scale input vectors by an amount comparable to their spectral norm (Claim 1). Single-example gradient updates are rank-one outer products, and for a rank-one matrix equality $\|\Delta Wh\| = \|\Delta W\|_{\text{op}} \|h\|$ holds

exactly (Claim 2). Combining these facts gives the scaling statements for features and their single-step changes in the paper (Equations (8)–(10)). 

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Conceptual-check questions (to test understanding)

I think these questions are maybe nicer than the premium version's because you are filling in the gaps left in the free version's proofs, so maybe you are building understanding more if you do the questions and read the free versions proof

1. Explain in words why a semi-orthogonal matrix with more rows than columns preserves the Euclidean norm of any input vector. Why does that immediately imply $\|Wh\| = \|W\|_{\text{op}}\|h\|$ in that case?
2. For W with i.i.d. $\mathcal{N}(0, \sigma^2)$ entries and fixed h , derive $\mathbb{E}\|Wh\|_2^2$. What does concentration of measure tell you about $\|Wh\|_2$ for large m ?
3. Give a short proof that for any vectors $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^n$, the operator norm of the rank-one matrix $ab^\top$ equals $\|a\|_2\|b\|_2$.
4. Suppose you use mini-batch gradient descent with batch size $B > 1$. Explain why ΔW may no longer be rank-one, and discuss whether you would still expect $\|\Delta Wh\|$ to be comparable to $\|\Delta W\|_{\text{op}}\|h\|$ in practice.
5. The paper uses “alignment” to argue lower bounds (so the upper bound from submultiplicativity is not too loose). Construct a simple example of a matrix W and vector h where $\|Wh\|$ is much smaller than $\|W\|_{\text{op}}\|h\|$. What property of random initialization prevents that example from being typical?
6. (Deeper) Using the Gaussian estimates above, show that if $m \gg n$, the ratio $\frac{\|Wh\|}{\|W\|_{\text{op}}\|h\|}$ tends to 1 (in probability). Sketch the argument and point out where each concentration result is used.

If you want, I can:

- Rewrite the short proofs above into a single polished paragraph suitable for inclusion in lecture notes.
- Produce numerical illustrations (quick Python experiments) showing $\|Wh\|$ vs $\|W\|_{\text{op}}\|h\|$ for semi-orthogonal and Gaussian initializations and for different aspect ratios m/n . (I can run and show code and plots if you want.)

Sources

I tried to get chatGPT to clarify it's argument by simplifying the presentation.. I felt maybe the explanations are a little convoluted

can you please simplify the explanations but also go through each step more thoroughly

Nice — I'll simplify and slow down the reasoning, and go through every algebraic step so nothing is hand-wavy. I keep the same two main claims (initialization alignment and rank-one update alignment), give short lemmas we use, then step through how those lemmas give the feature-scale and single-step change statements. I'll use plain English first and then show the algebraic steps where needed.

Quick roadmap

1. State the small list of facts we will use (proved or given).
2. Claim 1 (initialization): explain both the semi-orthogonal exact case and the Gaussian / random case step-by-step.
3. Claim 2 (update): show why single-example gradient updates are rank-one and why that produces an exact equality.
4. Put the two claims together to explain how forward feature norms and one-step feature changes scale.
5. Short extra notes about mini-batches and the “no perfect cancellation” assumption.
6. Conceptual check questions.

Facts we will use (stated plainly, proven when elementary)

Fact A — quadratic form identity. For any matrix W and vector h ,

$$\|Wh\|_2^2 = h^\top (W^\top W) h.$$

Why: $\|Wh\|_2^2 = (Wh)^\top (Wh) = h^\top W^\top Wh$.

Fact B — operator norm of a rank-one matrix. If $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^n$ then the matrix $A = ab^\top$ satisfies

$$\|A\|_{\text{op}} = \|a\|_2 \|b\|_2.$$

Short proof: For any x ,

$$Ax = a(b^\top x) \Rightarrow \|Ax\|_2 = |b^\top x| \|a\|_2 \leq \|b\|_2 \|x\|_2 \|a\|_2.$$

So $\|A\|_{\text{op}} \leq \|a\|_2 \|b\|_2$. To see equality is achievable, pick x parallel to b (i.e. $x = b/\|b\|_2$); then $|b^\top x| = \|b\|_2$ and $\|Ax\|_2 = \|a\|_2 \|b\|_2$. Thus $\|A\|_{\text{op}} = \|a\|_2 \|b\|_2$.

Fact C — simple Gaussian moment for a fixed vector. If each entry of $W \in \mathbb{R}^{m \times n}$ is i.i.d. $\mathcal{N}(0, \sigma^2)$ and $h \in \mathbb{R}^n$ is fixed, then each coordinate of Wh is Gaussian with variance $\sigma^2 \|h\|_2^2$, so

$$\mathbb{E}[\|Wh\|_2^2] = m \sigma^2 \|h\|_2^2.$$

(Concentration: when m is moderately large, $\|Wh\|_2^2$ is tightly concentrated around its expectation.)

Claim 1 — alignment at initialization

Goal: show $\|Wh\|_2$ is of the same order as $\|W\|_{\text{op}} \|h\|_2$ at initialization.

There are two versions: (1) **semi-orthogonal** initialization (exact), and (2) **Gaussian random** initialization (high-probability approximate).

(1) Semi-orthogonal initialization — exact and simple

Assumption. $W^\top W = I_n$ (this means the n columns of W are orthonormal; this is possible when $m \geq n$).

Some of the logic here is a little redundant, but this is a clearer explanation still.

Step-by-step:

- Using Fact A,

$$\|Wh\|_2^2 = h^\top (W^\top W)h = h^\top I_n h = h^\top h = \|h\|_2^2.$$

- Therefore $\|Wh\|_2 = \|h\|_2$.
- What is $\|W\|_{\text{op}}$ now? Because $W^\top W = I$, all singular values are ≥ 1 ? More precisely: the singular values of W are the square roots of eigenvalues of $W^\top W$, which are all 1, so all singular values equal 1. So $\|W\|_{\text{op}} = 1$.
- Combine: $\|Wh\| = \|h\| = \|W\|_{\text{op}} \|h\|$. This is an exact equality — no approximation.

Plain English intuition: A semi-orthogonal layer acts like a length-preserving map; it doesn't stretch or shrink any input vector — so the output length equals the input length, and that equals the spectral norm (which is 1) times the input length.

(2) Gaussian initialization — typical (probabilistic) alignment

Assumption. Entries W_{ij} are i.i.d. $\mathcal{N}(0, \sigma^2)$. Fix a nonzero input vector h .

Step 1 — size of $\|Wh\|$ on average.

Write the i -th coordinate of Wh as $(Wh)_i = \sum_{j=1}^n W_{ij} h_j$. Since each W_{ij} is mean zero Gaussian and h is fixed,

$$\text{Var}((Wh)_i) = \sigma^2 \sum_{j=1}^n h_j^2 = \sigma^2 \|h\|_2^2.$$

So $(Wh)_i \sim \mathcal{N}(0, \sigma^2 \|h\|_2^2)$. Because the m rows are independent,

$$\mathbb{E}[\|Wh\|_2^2] = m \sigma^2 \|h\|_2^2.$$

Thus on average $\|Wh\|_2 \approx \sigma \sqrt{m} \|h\|_2$.

Step 2 — typical size of $\|W\|_{\text{op}}$.

For a random Gaussian matrix, standard non-asymptotic random matrix facts (well-known; you can find proofs in many texts) say

$$\|W\|_{\text{op}} \text{ is about } \sigma(\sqrt{m} + \sqrt{n}).$$

Interpretation: the operator norm is somewhat larger than the typical singular value $\sigma \sqrt{m}$ by an additive $\sigma \sqrt{n}$ term.

Step 3 — compare the two.

Take the ratio

$$\frac{\|Wh\|_2}{\|W\|_{\text{op}} \|h\|_2} \approx \frac{\sigma \sqrt{m}}{\sigma(\sqrt{m} + \sqrt{n})} = \frac{\sqrt{m}}{\sqrt{m} + \sqrt{n}}.$$

I appreciated ChatGPT taking the time to establish this, rather than just citing it. It is probably something a student in CS182 could figure out, but I think it's still better to present it than not.

If the layer is **wide** so that $m \geq n$, then $\sqrt{m}/(\sqrt{m} + \sqrt{n}) \geq 1/2$. So $\|Wh\|$ is at least a constant fraction of $\|W\|_{\text{op}}\|h\|$. In words: the random input doesn't get projected purely into tiny singular directions with overwhelming probability — the output norm is the same order as the spectral norm times $\|h\|$.

Plain English intuition: A Gaussian matrix has many singular directions of comparable size when the matrix is wide; a fixed input vector will typically line up enough with these directions that its amplification is roughly the same order as the maximum amplification the matrix can produce.

Claim 2 — exact alignment for a single-example gradient update

Goal: show that for a single training example, the gradient update ΔW is rank-one and $\|\Delta Wh\| = \|\Delta W\|_{\text{op}}\|h\|$ for that same input h .

Why ΔW is rank-one (backprop truth)

Consider a fully connected layer that computes $z = Wh_{\text{in}}$ and then some scalar loss L depends (possibly through later layers) on z . By the chain rule,

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial z} \frac{\partial z}{\partial W}.$$

But $\partial z / \partial W$ is exactly $h_{\text{in}}^\top$ in matrix form: for each weight W_{ij} , the derivative of z_i w.r.t. W_{ij} is $h_{\text{in},j}$. Thus for a single training example the gradient has the structure

$$\Delta W = -\eta \cdot (g) h_{\text{in}}^\top,$$

where $g \equiv \partial L / \partial z$ is a vector (usually called the backpropagated error) and η is the learning rate (sign included depending on convention). This is an outer product $ab^\top$ with $a = g$ and $b = h_{\text{in}}$; therefore ΔW is rank-one.

Why we get exact equality for $\|\Delta Wh_{\text{in}}\|$

Presentation here is also much clearer, though maybe it could have cited the facts it previously established (eg. by fact 1 ...)

Apply Fact B (operator norm of a rank-one matrix). For $A = ab^\top$,

$$\|A\|_{\text{op}} = \|a\| \|b\|.$$

Now compute Ab :

$$Ab = a(b^\top b) = a \|b\|^2.$$

So

$$\|Ab\| = \|a\| \|b\|^2 = (\|a\| \|b\|) \|b\| = \|A\|_{\text{op}} \|b\|.$$

Replace $a = g$ and $b = h_{\text{in}}$ and $A = \Delta W$. We get

$$\|\Delta Wh_{\text{in}}\|_2 = \|\Delta W\|_{\text{op}} \|h_{\text{in}}\|_2.$$

This is an exact identity — no probability or approximation.

Plain English intuition: a rank-one matrix maps the particular vector that appears in its second factor (here h_{in}) to a multiple of the first factor. The ratio of output norm to input norm equals the matrix spectral norm for that vector exactly.

Putting the claims together: how a single SGD step changes a hidden feature

I think this is a better “tying in” section than previously too.

We want the size of the change in the hidden activations at layer ℓ , Δh_ℓ , after a small weight update. Take the first-order expansion (linearization) of how the output $h_\ell = W_\ell h_{\ell-1}$ changes when weights and earlier activations change:

$$\Delta h_\ell \approx \underbrace{\Delta W_\ell h_{\ell-1}}_{\text{(A) direct weight effect}} + \underbrace{W_\ell \Delta h_{\ell-1}}_{\text{(B) propagated earlier changes}} + \text{(C) higher order } (\Delta W_\ell \Delta h_{\ell-1}).$$

- Term **(A)** we just handled exactly using Claim 2 for single-example SGD:

$$\|\Delta W_\ell h_{\ell-1}\| = \|\Delta W_\ell\|_{\text{op}} \|h_{\ell-1}\|.$$

So the *direct* change at layer ℓ due to updating weights at that layer scales like $\|\Delta W_\ell\|_{\text{op}} \|h_{\ell-1}\|$.

- Term **(B)** is the pre-existing weight W_ℓ applied to the change in the previous layer's activations $\Delta h_{\ell-1}$. Using Claim 1 about weight scale we can say roughly

$$\|W_\ell \Delta h_{\ell-1}\| \lesssim \|W_\ell\|_{\text{op}} \|\Delta h_{\ell-1}\|.$$

This is a propagation of earlier changes; it will be important when we consider the whole network, but it is typically not larger than (A) if the update is small and not adversarially canceled.

- Term **(C)** is second order in small quantities — the product of two small changes — so for small learning rate it is negligible compared to (A).

Conclusion (informal): so long as (A) does not exactly cancel (B) + (C), the magnitude $\|\Delta h_\ell\|$ is comparable to $\|\Delta W_\ell\|_{\text{op}} \|h_{\ell-1}\|$. If we use Claim 1 to replace $\|h_{\ell-1}\|$ by $\|W_{\ell-1}\|_{\text{op}} \|h_{\ell-2}\|$ etc., we get the scaling statements the paper writes down: both the forward feature norm and its single-step change scale like the spectral norm times the input norm.

Extra clarifications and small practical points

1. **Mini-batch updates (batch size $B > 1$).** The gradient for a batch is the sum of B outer products (one per example), so ΔW is the sum of up to B rank-one matrices (rank at most B). Fact B no longer gives an exact equality for $\|\Delta W h\|$, but in practice we still expect $\|\Delta W h\|$ to be comparable to $\|\Delta W\|_{\text{op}} \|h\|$ unless the different gradients cancel each other in a finely tuned way.
2. **“No perfect cancellation” assumption.** This is a genericity assumption: we assume the directions of the various terms are not arranged to produce precise vector cancellation. In high dimensions, exact cancellation is unlikely except in contrived examples.

3. **Why width matters in the random case.** If the layer is narrow (small m), the largest singular value can be much larger than the *typical* amplification, so a carefully chosen input could produce $\|Wh\|$ much smaller than $\|W\|_{\text{op}}\|h\|$. Random initialization avoids this typical worst-case alignment: a random h will *not* align only with tiny singular directions.

Very short numeric intuition (toy numbers)

- Suppose Gaussian init with $\sigma = 1$, $m = 1000$, $n = 500$.
 - Typical $\|Wh\| \approx \sqrt{m}\|h\| \approx 31.6\|h\|$.
 - Typical $\|W\|_{\text{op}} \approx \sqrt{m} + \sqrt{n} \approx 31.6 + 22.4 \approx 54$.
 - Ratio $\approx 31.6/54 \approx 0.585$ — same order (not tiny).
- Semi-orthogonal: $\|Wh\| = \|h\|$ and $\|W\|_{\text{op}} = 1$ so ratio = 1 exactly.

Conceptual check questions (practice)

1. Prove directly (using matrix algebra) that if $W^T W = I$ then $\|Wh\| = \|h\|$ for all h . What does this say about singular values of W ?
2. For a Gaussian matrix W with i.i.d. $\mathcal{N}(0, \sigma^2)$ entries and fixed vector h , compute $\mathbb{E}[\|Wh\|^2]$. Why does the law of large numbers tell you $\|Wh\|$ concentrates around that value for large m ?
3. Give the short proof that $\|ab^T\|_{\text{op}} = \|a\| \|b\|$. Then show algebraically why $ab^T b = \|b\|^2 a$, and combine those to prove $\|ab^T b\| = \|ab^T\|_{\text{op}} \|b\|$.
4. Suppose you run SGD with batch size B . Write ΔW as a sum of outer products and explain why ΔW may now have rank up to B . Is the exact equality $\|\Delta Wh\| = \|\Delta W\|_{\text{op}}\|h\|$ still guaranteed? Why or why not?
5. Construct a simple example (explicit W and h) where $\|Wh\|$ is much smaller than $\|W\|_{\text{op}}\|h\|$. Then explain why that example is unlikely under a wide Gaussian initialization.
6. In the expansion $\Delta h_\ell = \Delta W_\ell h_{\ell-1} + W_\ell \Delta h_{\ell-1} + \Delta W_\ell \Delta h_{\ell-1}$, suppose $\|\Delta W_\ell\|_{\text{op}} = \epsilon$ is small and $\|W_\ell\|_{\text{op}} = 1$. Give bounds (in ϵ) for the sizes of the three terms if $\|\Delta h_{\ell-1}\| = O(\epsilon)$. Which term dominates for small ϵ ?

If you'd like, I can now:

- Convert the algebraic steps into a one-page “cheat sheet” with only the minimal equations you'd paste into lecture notes, or
- Run a tiny Python experiment (and show numbers) that generates random W , a fixed h , and prints $\|Wh\|$, $\|W\|_{\text{op}}\|h\|$, and their ratio for different aspect ratios m/n . Which would you prefer?